

ROTATION-EQUIVARIANT MULTI-SCALE CONVOLUTION VIA ADAPTIVE MAGNITUDE LBP

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ABSTRACT

Rotation equivariance is a key challenge for deep neural networks, as standard convolutions are sensitive to input orientation and often rely on extensive data augmentation. We introduce MLBPConv, a novel rotation-equivariant convolution based on an adaptive magnitude-thresholded Local Binary Pattern. Specifically, we first identify that the instability of prior LBP-based methods stems from the intrinsic periodicity of LBP codes, which can cause errors in orientation alignment, and show that MLBPConv effectively alleviates this issue. To further enrich feature representation, we design a lightweight multi-scale fusion strategy that aggregates multi-receptive-field features without increasing the number of learnable parameters. Experiments on MNIST-Rot, Outex_TC.00012, and MTARSI datasets demonstrate that our method outperforms standard CNNs and classical rotation-equivariant networks, and integrating MLBPConv into ResNet architectures substantially enhances rotation robustness. This approach provides an efficient, lightweight, and plug-and-play solution for learning stable rotation-equivariant features.

Index Terms— Rotation Equivariance, Local Binary Pattern, Multi-Scale Feature Fusion, Lightweight Networks

1. INTRODUCTION

Geometric equivariance of image features refers to the property that when an input image undergoes a certain geometric transformation, its feature representation changes in a consistent and predictable manner, thereby preserving structural relationships and semantic information. Since 2012, research on geometric equivariance in computer vision has gradually shifted from the design of handcrafted features to the construction of geometric-equivariant deep neural networks (DNNs), with a particular focus on how to incorporate equivariance into network architectures—especially into convolution, the core operation of DNNs. Conventional convolution is only translation-equivariant and is therefore limited in handling other geometric transformations. In recent years,

increasing attention has been devoted to extending equivariance to rotational transformations. Existing studies have demonstrated that rotation-equivariant convolutions exhibit significant potential in applications such as biomedical image classification [1], multi-orientation object detection [2], remote sensing image semantic segmentation [3], image generation [4] and so on.

Early attempts to achieve rotation equivariance in convolution primarily relied on data augmentation during training, i.e., randomly rotating input images by certain angles. However, the rotation equivariance obtained in this manner is not generalizable. Taking classification tasks as an example, the network only exhibits equivariance for categories that are augmented during training, and this property cannot be extended to categories that have not undergone augmentation. Moreover, extensive data augmentation may introduce substantial parameter redundancy in the network [5]. Consequently, later research focused on embedding explicit rotational equivariance into network architectures by redefining convolution operators. Methods such as G-CNN [6] and E2CNN [7] employ group convolutions to achieve rotation equivariance, while RotEqNet [8] and ORN [9] rotate convolutional kernels to extract multi-orientation features. Nevertheless, these approaches often differ from standard convolutions in parameter count and computational flow, making them difficult to integrate as plug-and-play modules into mainstream backbones like ResNet. They also typically guarantee strict equivariance only for discrete rotation angles, e.g., multiples of 90°.

The newest studies have explored the use of non-learnable operators to achieve rotation equivariance in convolution, such as Sobel operators [10], sorting operations [11], and rotation-invariant coordinate systems [12]. These methods were originally primarily employed to construct handcrafted features with rotation equivariance or invariance. Compared with group convolution-based networks such as G-CNN, their main advantage lies in the fact that the convolutional structure maintains the same number of learnable parameters and computational procedures as standard convolution, allowing them to be used as plug-and-play modules within existing neural network architectures. In addition, these methods are generally capable of handling arbitrary rotation angles. A

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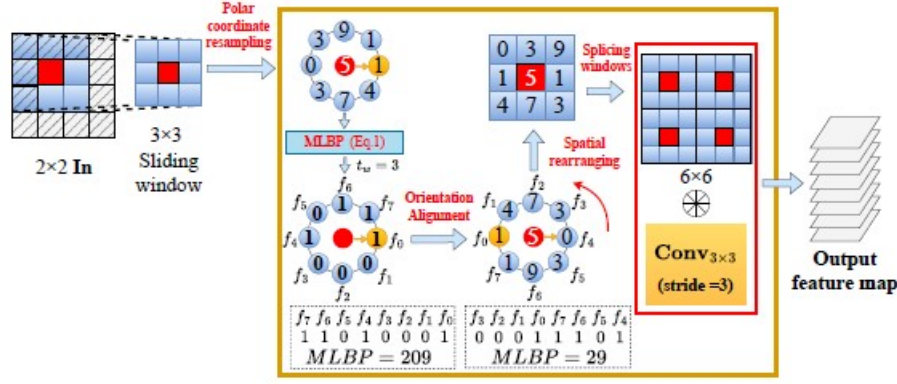


Fig. 1. Illustration of the MLBPConv pipeline with a 3×3 convolutional kernel.

representative prior work is the Regional Rotate Layer (RRL) proposed by Hao et al. [13], which aligns the orientation of local neighborhoods using the most famous textural image descriptor, Local Binary Pattern (LBP) [14], to achieve rotation equivariance of the following standard convolution operation. However, experiments show that RRL’s rotation equivariance is considerably unstable, severely limiting its practical applicability.

In this paper, we propose MLBPConv, a novel rotation-equivariant convolution based on a LBP variant with adaptive magnitude threshold. We reveal, for the first time, that the inherent periodicity of LBP causes the instability of RRL and similar prior methods, and show that MLBPConv effectively addresses this issue. To capture both fine-grained textures and broader structural information efficiently, we further introduce a lightweight multi-scale feature fusion strategy. The main contributions of this work are as follows:

1) We propose rotation-equivariant MLBPConv based on an LBP variant with an adaptive magnitude threshold. We show that the inherent periodicity of LBP causes rotation angle estimation errors in representative prior methods such as RRL, and statistical analysis confirms that MLBPConv resolves this instability to some extent.

2) We design a lightweight multi-scale feature extraction strategy using MLBPConv, achieving multi-scale representation while maintaining the same number of learnable parameters as single-scale convolution.

3) Comprehensive experiment results show that MLBPConv’s rotation equivariance is more stable than RRL and, when integrated into classic network architectures with varying depths, improves classification on Outex_TC_00012 and MTARSI datasets by 16.2% and 4.1% over the baseline, surpassing RRL by 5.5% and 4.1%, with the lightweight multi-scale MLBPConv providing additional gains.

2. METHODOLOGY

This section proposes a novel rotation-equivariant convolution, MLBPConv, based on a magnitude-thresholded LBP operator. We first reveal a critical flaw in the equivariance sta-

bility of representative prior methods and show how MLBPConv addresses it. To efficiently capture both fine-grained textures and broader structural information, we further design a lightweight multi-scale fusion strategy for MLBPConv.

2.1. Pipeline of Rotation-Equivariant MLBPConv

As an example, Figure 1 illustrates the pipeline of the proposed MLBPConv with a 3×3 kernel. For an input feature map $\mathbf{In} \in \mathbb{R}^{H \times W}$, we pad it with 1 ring of zeros, and extract 3×3 sliding windows with a stride of 1. For each window, a polar coordinate system (r, θ) is established with the origin at the center for resampling, and feature values f_i at sample points $(1, \theta_i)$ are obtained via bilinear interpolation, where $\theta_i = 2\pi i/8, i = 0, 1, \dots, 7$. The point directly to the right of the center is selected as the initial principal direction.

We compute an LBP variant based on an adaptive magnitude threshold, termed MLBP, on the obtained feature values f_i . First, the magnitude difference $m_i = |f_i - f_c|$ between each neighbor and the center point is calculated, where f_c is the feature value at the center. Then, the mean of these magnitudes serves as an adaptive threshold $t_m = \frac{1}{8} \sum_{i=0}^7 m_i$ for binarization:

$$MLBP = \sum_{i=0}^7 s(m_i \geq t_m) \cdot 2^i \quad (1)$$

where $s(x)$ equals 1 if x is true, and 0 otherwise.

MLBP code can be used to align the convolutional neighborhood to a canonical orientation, thereby eliminating the effect of rotation angles. Specifically, we circularly shift MLBP code and select the minimum value $MLBP^*$. Apply the circular shift that yields $MLBP^*$ to process $f_0 \sim f_7$, and then place the reordered feature values into the 8-neighborhood of the center in a left-to-right, top-to-bottom order. According to their spatial positions, all aligned 3×3 sliding windows are stitched together to construct a new feature map of size $(3 \cdot H) \times (3 \cdot W)$. Then, we perform a 3×3 convolution with a stride of 3 on this new input.

Since rotation of the input $\mathbf{In}$ induces only a circular permutation of the sampled neighbor values $f_0 \sim f_7$, and

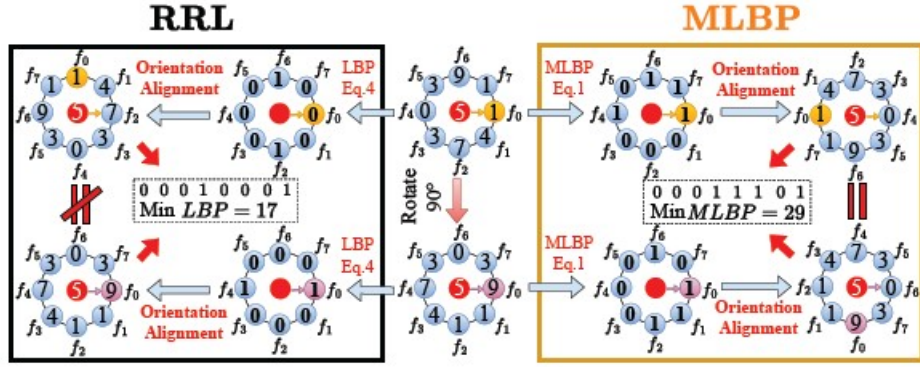


Fig. 2. Comparison of cyclic ambiguity in orientation alignment between RRL (using original LBP) and the proposed MLBP, highlighting the stability improvement of MLBP.

MLBP aligns each local neighborhood to a unique canonical orientation via minimum circular encoding, rotated versions of the same local structure are consistently normalized before convolution. Therefore, the resulting feature maps computed from a rotated input are correspondingly rotated versions of those from the original input, i.e., MLBPConv satisfies the rotation equivariance property. Obviously, the resulting output has a spatial size of $H \times W$, which is identical to the output size obtained by applying a standard 3×3 convolution with stride 1 directly to the original input $\mathbf{In}$. Moreover, since the entire MLBP computation involves no learnable parameters, MLBPConv has the same number of learnable parameters as standard convolution. Therefore, the two can be used interchangeably.

Building upon this property, consider a classical convolutional neural network composed of L convolutional layers. If each standard convolution is replaced by its corresponding MLBPConv, the resulting network $\Phi(\cdot)$ becomes a composition of rotation-equivariant operators:

$$\Phi(\mathbf{In}) = \mathcal{M}_L(\mathcal{M}_{L-1}(\cdots \mathcal{M}_1(\mathbf{In}))), \quad (2)$$

where each $\mathcal{M}_l$ represents an MLBPConv layer. Since rotation equivariance is preserved under composition, for any rotation operator $\mathcal{R}_\phi$, we have

$$\Phi(\mathcal{R}_\phi(\mathbf{In})) = \mathcal{R}_\phi(\Phi(\mathbf{In})). \quad (3)$$

which means that the final feature maps of $\Phi(\cdot)$ remain rotation-equivariant. If a spatial pooling operation (e.g., average pooling) is applied to reduce the feature maps to a 1×1 spatial resolution, the rotational transformation no longer affects the output values, yielding rotation-invariant feature representations. These invariant features can then be fed into classification or regression heads for downstream tasks.

2.2. The Equivariance Stability: MLBPConv vs. RRL

As mentioned earlier, Hao et al. also employed the LBP operator to achieve rotation equivariance in convolution and proposed RRL [13]. Different from MLBP, they utilize the orig-

inal LBP to perform orientation alignment of the local neighborhood:

$$LBP = \sum_{i=0}^7 s(f_i \geq f_c) \cdot 2^i \quad (4)$$

However, experiments show that RRL's equivariance is not stable, which limits its practical utility, and the authors did not identify the underlying cause. In fact, this instability arises from inherent periodicity of LBP code. As shown in Figure 2, the minimal LBP of the 3×3 neighborhood before and after a 90° is both 00010001. However, since the pattern "0001" appears twice, such cyclic codes can result in ambiguity in determining the neighborhood reordering, as multiple circular shift schemes may yield the same minimum LBP pattern during orientation alignment. This problem is especially pronounced in binary or low-texture images, where local patterns lack sufficient distinctive details.

In contrast, MLBP uses magnitude-based thresholding, which suppresses small fluctuations and retains only significant local variations, producing sparser and more irregular codes, thereby reducing cyclic patterns and improving stability during orientation alignment. Our statistics on the Outex_TC_00012 texture dataset show that the probability of generating cyclic codes is 13.1% for the original LBP, whereas it is only 5.7% for MLBP, representing a 56.49% reduction. This significantly decreases cyclic ambiguity and ensures more stable rotation equivariance.

2.3. Lightweight Multi-Scale MLBPConv

Multi-scale feature fusion enhances representation by capturing both fine-grained textures and broader structures. However, naively increasing the convolution kernel size for larger scales would increase the number of learnable parameters, expanding the parameter space and making the network harder to train or converge. In this paper, we design a lightweight strategy of multi-scale MLBPConv feature extraction and fusion.

As shown in Figure 3, when computing MLBPConv over a 7×7 neighborhood, circular neighborhoods with radii $r =$

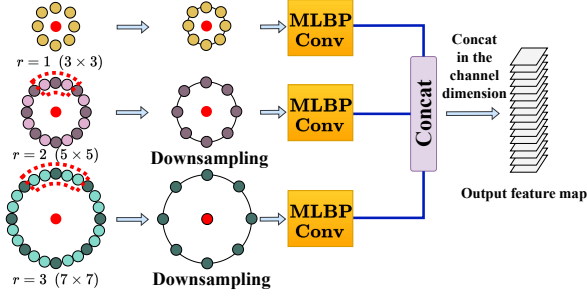


Fig. 3. The proposed lightweight multi-scale MLBPConv^{ms} .

$\{1, 2, 3\}$ are first sampled using 8, 16, and 24 points, respectively. The sampled values are then averaged to downsample larger neighborhoods into a standardized 8-neighborhood:

$$f_i = \frac{1}{r} \sum_{j=0}^{r-1} In(r, \theta_{ir+j}), \quad i = 0, \dots, 7, \quad (5)$$

This allows neighborhoods with different radii to be uniformly represented as 3×3 neighborhoods consisting of eight sampled points and one center point, enabling the use of the same 3×3 convolution operation. The feature maps obtained from different radii, denoted as $\text{Out}_1, \text{Out}_2, \text{Out}_3$, are then concatenated along the channel dimension:

$$\text{MLBPConv}^{ms} = \text{Concat}(\text{Out}_1, \text{Out}_2, \text{Out}_3) \quad (6)$$

where the superscript ms indicates multi-scale MLBPConv.

Assuming that the total number of output channels after concatenating multi-scale features is equal to that of the original 7×7 convolution, the overall parameter count is determined by three parallel 3×3 convolutions. Denoting C_{in} and C_{out} as the input and output channels, the parameter ratio compared with a standard 7×7 convolution is:

$$\text{Ratio} = \frac{3^2 \cdot C_{in} \cdot C_{out}}{7^2 \cdot C_{in} \cdot C_{out}} = \frac{9}{49} \approx 0.184. \quad (7)$$

This demonstrates that the proposed multi-scale MLBPConv efficiently integrates information across scales while maintaining a lightweight parameter footprint.

3. EXPERIMENTS

3.1. Datasets and Experimental Setup

We evaluate our method on three datasets: **(1) MNIST** [15]: This dataset consists of 70000 28×28 grayscale handwritten digit images (0~9). The original dataset contains 60000 training and 10000 testing images. We randomly selected 10000 images from the training set for validation, leaving the remaining 50000 for training. The MNIST-Rot test set was constructed by rotating each of the 10000 original test images at 36 angles ($0^\circ, 10^\circ, \dots, 350^\circ$), resulting in 360000 rotated samples. **(2) Outex_TC_00012** [16]: This texture dataset

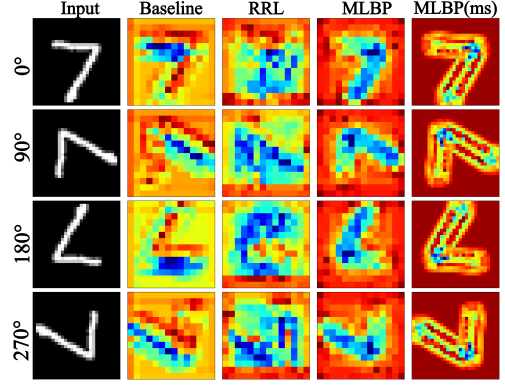


Fig. 4. Feature heatmap visualization of digit “7” under various rotations, illustrating the rotation equivariance of the proposed MLBPConv and MLBPConv^{ms} .

contains 9120 128×128 grayscale images across 24 classes. The training set (1440 images) includes 20 surfaces per class captured under three illumination conditions (inca, t184, horizon). The test set (7680 images) comprises the same surfaces under t184 and horizon illuminations at 8 distinct rotation angles (5° to 90°). **(3) MTARSI** [17]: This dataset contains 9385 128×128 RGB aircraft images spanning 20 categories. We randomly selected 100 images per class for training (2000 total), with the remaining 7385 images serving as the test set. The test images naturally exhibit significant rotational variations due to diverse aerial viewpoints.

All experiments were conducted on an NVIDIA RTX 4090 GPU (24GB) with Ubuntu 22.04 and PyTorch 2.5.1. All models were trained from scratch without data augmentation. For MNIST-Rot, we used Adam with an initial learning rate of 1×10^{-4} , decayed by 0.8 every 10 epochs, for 100 epochs and a batch size of 100. For Outex_TC_00012 and MTARSI, Adam was used with a learning rate of 1×10^{-3} , decayed by 0.6 every 10 epochs, for 100 epochs and a batch size of 10.

3.2. Rotation Equivariance Evaluation of MLBPConv

We first evaluate the rotation equivariance of MLBPConv on the MNIST-Rot dataset. The baseline model consists of six 3×3 convolutional layers with output channels of 32, 32, 64, 64, 128, and 128. A 2×2 max-pooling layer follows the second and fourth convolutions, and a 7×7 average pooling layer and a fully connected layer are applied at the end. By replacing all 3×3 convolutions with MLBPConv, we obtain the corresponding rotation-equivariant model. Furthermore, in the first 64-channel convolutional layer, 4 channels are replaced by 5×5 MLBPConv (radius $r = 2$) and 4 channels by 7×7 MLBPConv (radius $r = 3$), forming the multi-scale MLBPConv^{ms} model. For comparison, we also construct rotation-equivariant counterparts by replacing the standard convolutions with RRL [13], and include other classical rotation-equivariant networks such as ORN [9], RotEqNet

Table 1. Classification accuracy on original MNIST test set and MNIST-Rot test set.

Methods	Input Size	MNIST	MNIST-Rot
ORN[9]	32×32	99.42%	80.01%
RotEqNet[8]	28×28	99.26%	73.20%
G-CNN[6]	28×28	99.27%	44.81%
Baseline	28×28	99.45%	45.48%
RRL[13]	28×28	99.27%	44.98%
MLBPConv	28×28	98.63%	80.90%
MLBPConv ^{ms}	28×28	98.25%	83.75%

[8], and G-CNN [6].

As shown in Table 1, all methods achieve comparable accuracy on the original MNIST test set. However, on MNIST-Rot with significant rotational variations, MLBPConv improves accuracy by 35.4% and 35.9% over the Baseline and RRL, respectively. Notably, RRL performs even worse than the Baseline, as LBP codes computed on binary images are highly prone to periodic patterns, which severely disrupt rotation equivariance and lead to unstable performance. In contrast, using MLBP for orientation alignment effectively alleviates this issue. Moreover, MLBPConv^{ms} achieves the best performance on MNIST-Rot, further improving over MLBPConv by 2.85%, validating the effectiveness of the multi-scale fusion strategy. Feature visualizations in Figure 4 further confirm that MLBPConv and MLBPConv^{ms} exhibit clear rotation equivariance, whereas the Baseline and RRL do not.

3.3. MLBPConv Integrated with Common DNN Models

To demonstrate that MLBPConv can serve as a plug-and-play module in classical DNN architectures, we adopt ResNet18, ResNet34, and ResNet50 as baselines. All non-1 $\times$ 1 convolutional layers are replaced with MLBPConv, yielding the corresponding rotation-equivariant models. To evaluate the multi-scale variant, we further modify the first 3×3 convolutional layer (64 channels) by allocating 4 channels to 5×5 MLBPConv (radius $r = 2$) and 4 channels to 7×7 MLBPConv (radius $r = 3$), forming the MLBPConv^{ms} model. For fair comparison, we also replace the standard convolutions with RRL [13] to construct the corresponding rotation-equivariant counterparts.

Table 2 presents the classification performance of our methods on the Outex_TC_00012 and MTARSI datasets. It can be observed that both MLBPConv and MLBPConv^{ms} consistently outperform the Baseline and RRL across various ResNet architectures. On the Outex_TC_00012 dataset, MLBPConv improves the original ResNet models by an average of 16.2%, while achieving a 4.1% improvement on the MTARSI dataset. These performance gains are obtained

Table 2. Classification accuracy on Outex_TC_00012 dataset and MTARSI dataset.

Methods	Outex_TC_00012	MTARSI	Para.
ResNet18	65.06%	88.59%	11.18M
RRL[13]	75.11%	89.05%	11.18M
MLBPConv	81.46%	93.15%	11.18M
MLBPConv ^{ms}	79.79%	92.34%	11.18M
ResNet34	63.21%	87.41%	21.29M
RRL[13]	75.82%	88.70%	21.29M
MLBPConv	80.20%	90.64%	21.29M
MLBPConv ^{ms}	85.87%	91.24%	21.29M
ResNet50	65.91%	87.44%	23.54M
RRL[13]	75.53%	85.55%	23.54M
MLBPConv	81.19%	91.89%	23.54M
MLBPConv ^{ms}	81.13%	90.94%	23.54M

without increasing the number of learnable parameters. Due to the strong representation and semantic modeling capacity of ResNet, the Baseline already achieves high performance, leading to smaller marginal gains from the rotation invariance of MLBPConv compared with the shallow model in Section 3.2. Similarly, since MTARSI contains richer semantic information, its performance improvement is less pronounced than on the texture-dominated Outex_TC_00012 dataset. Moreover, given the limited dataset scale, ResNet50 is more prone to overfitting, while ResNet18 is better matched to the data size. Notably, MLBPConv^{ms} performs best on ResNet34, suggesting that a moderate network depth achieves a better balance between representation capacity and feature diversity, allowing multi-scale information to be more effectively exploited. In contrast, ResNet18 has limited capacity, whereas ResNet50 may suffer from overfitting, reducing the additional gains. Overall, MLBPConv and MLBPConv^{ms} demonstrate strong generalization across different network depths.

4. CONCLUSION

In summary, we propose MLBPConv, a novel rotation-equivariant convolution based on an LBP variant with an adaptive magnitude threshold, which effectively resolves the instability caused by the inherent periodicity of LBP in prior methods such as RRL. By further introducing a lightweight multi-scale feature fusion strategy, our approach captures both fine-grained textures and broader structural information without increasing learnable parameter count. Extensive experiments demonstrate that MLBPConv consistently achieves stable rotation equivariance and significantly improves classification performance across different network architectures and datasets, with the multi-scale variant providing additional gains, confirming its efficiency, robustness, and general applicability.

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